

Machine Learning

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1 Problems

P 1. Find the orthogonal projection of a vector v onto a plane $\mathcal{P}$ described by the equation $\theta \cdot x + \theta_0 = 0$.

The component of v orthogonal to $\mathcal{P}$ is $\frac{v \cdot \theta}{\|\theta\|^2} \theta$. The component of v parallel to $\mathcal{P}$ is $w = v - \frac{v \cdot \theta}{\|\theta\|^2} \theta$. Consider the vector $v - \frac{v \cdot \theta + \theta_0}{\|\theta\|^2} \theta$, which is on the line vw . This point lies on $\mathcal{P}$ and has the same component parallel to $\mathcal{P}$ as v . Hence, it must be the orthogonal projection of v onto $\mathcal{P}$.

P 2. Gaussian distribution. Define $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)}$. Show that $\int_{-\infty}^{\infty} f_X(x) dx = 1$. Let $y = (x - \mu)/(\sigma\sqrt{2})$. Then $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-y^2}$ and

$$\begin{aligned} \int_{-\infty}^{\infty} f_X(x) dx &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} e^{-y^2} \sigma\sqrt{2} dy \\ &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-y^2} dy \\ &= \frac{1}{\sqrt{\pi}} \sqrt{\pi} \\ &= 1. \end{aligned}$$

As a corollary, the limit of the integral is 1 as $\sigma \rightarrow 0$.

Perceptron Algorithm Objective function.

$$J(\theta, \theta_0) = \frac{1}{n} \sum_{i=1}^n \text{Loss}_h(y^{(i)}(\theta \cdot x^{(i)} + \theta_0)) + \frac{\lambda}{2} \|\theta\|^2.$$

λ is the regularization parameter, $\lambda > 0$. The validation set is taken to be a subset of the training set to simulate a curve for c^* , for which c^* can be estimated.